

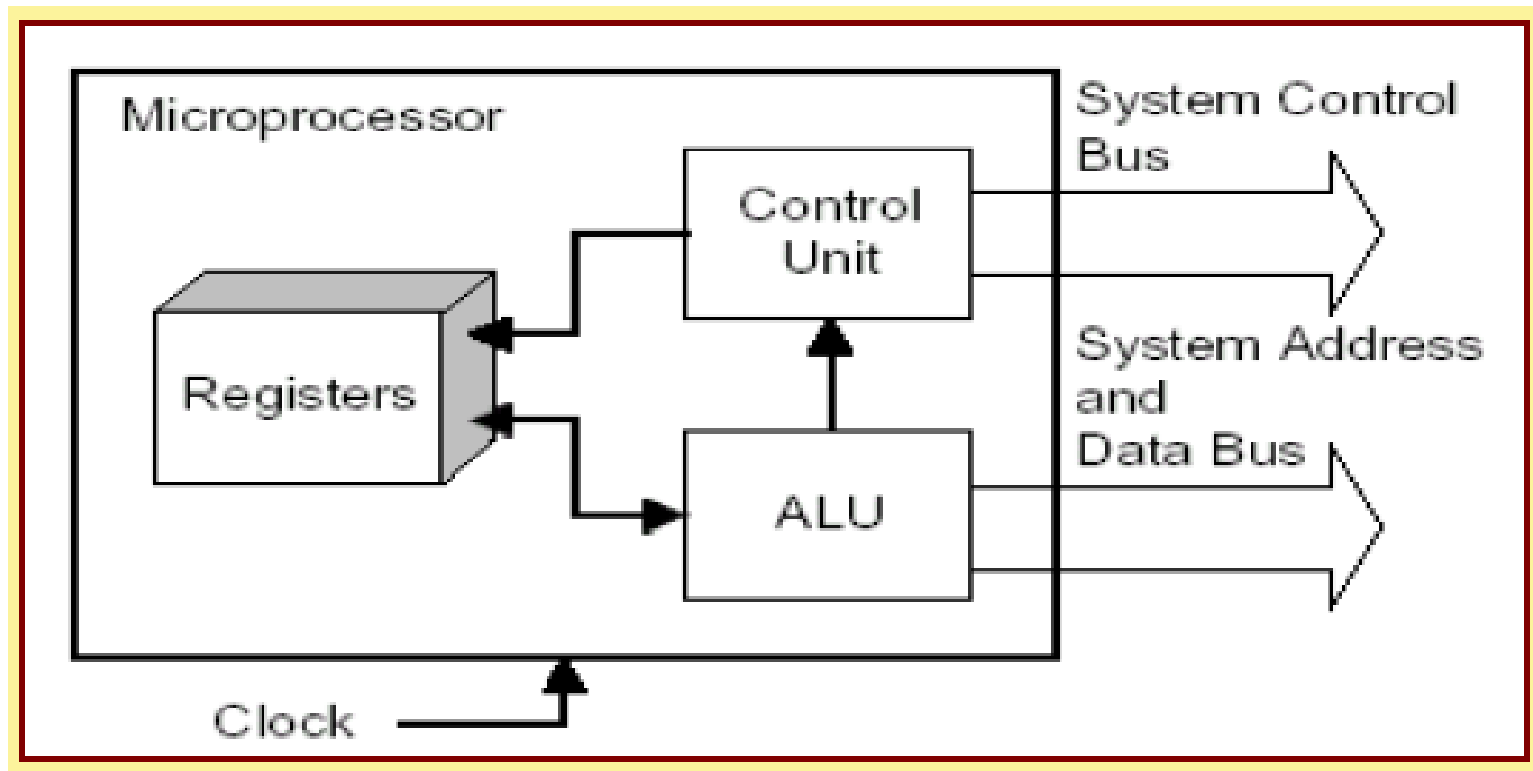
ARM Processor Architecture

ANUSHA N

Some Basics

What is a Microprocessor?

It is basically the entire CPU fabricated on a single chip.

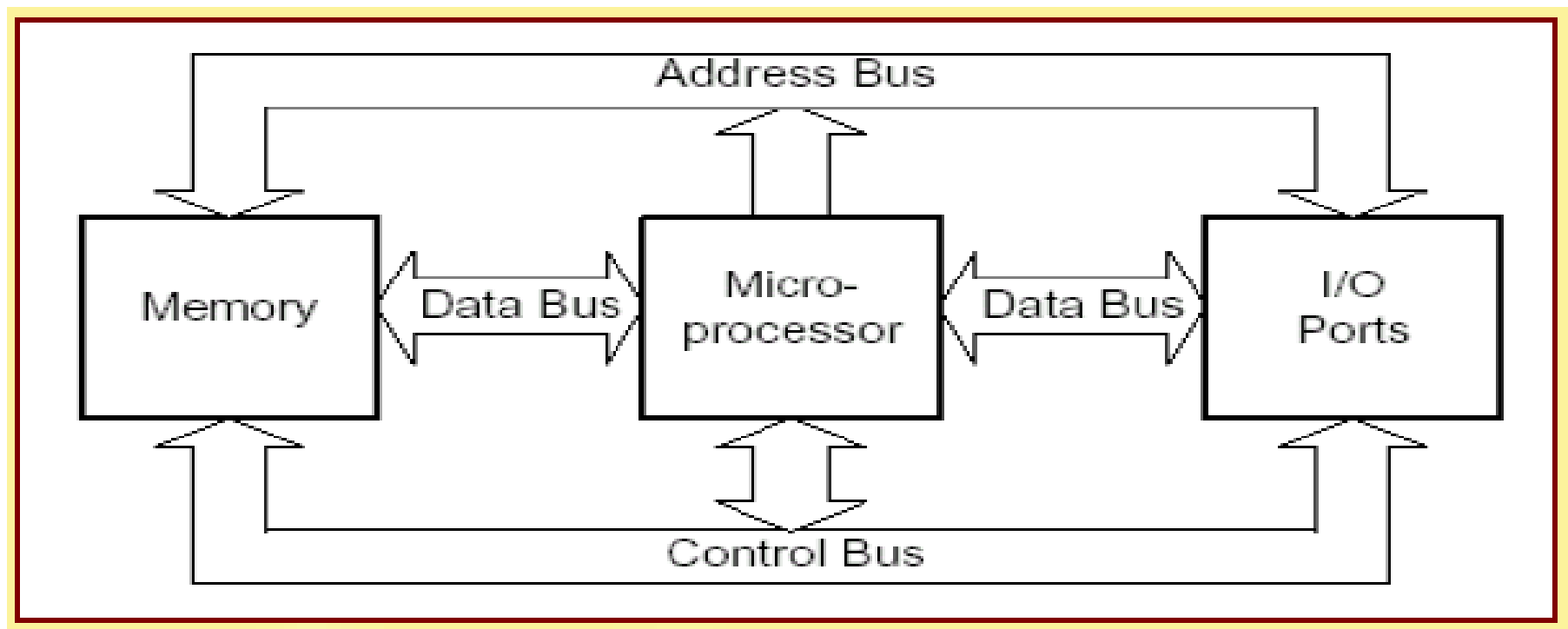


Some Basics

What is a Microcomputer?

It is a computer system built using a microprocessor.

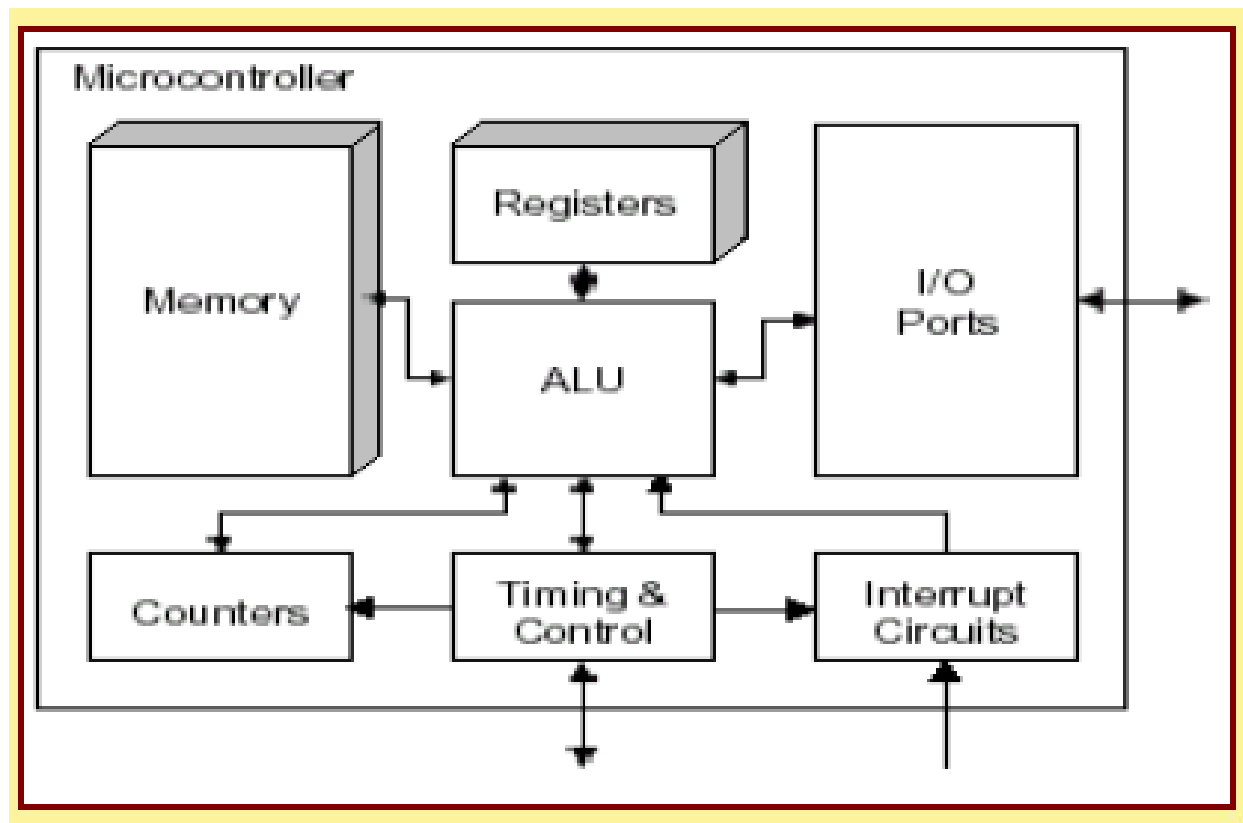
- Too complex and expensive for very small and low-cost embedded systems.



Some Basics

Microcontrollers: The Heart of Embedded Systems

- Very inexpensive, small, low power.
- Convenient for use in embedded system design.



Why do we talk about ARM?

- One of the most widely used processor cores.
- Some application examples:
 - ARM7: iPod
 - ARM9: BenQ, Sony Ericsson
 - ARM11: Apple iPhone, Nokia N93, N100
- 90% of 32-bit embedded RISC processors till 2010.
- Mainly used in battery-operated devices:
- Due to low power consumption and reasonably good performance.

ARM Powered Applications

Samsung ML5100A



Diamond Multimedia Rio 600



JVC "Pixstar" GC-X1



Alba Bush Internet TV



3Com 10/100 PCI NIC



Lexmark Z52 Color Jetprinter



Nintendo Gameboy Advance



Iomega HipZip



Sony MZ-R90 MiniDisc



Ericsson R380



HP CapShare



HP Jornada 820



Psion Revo Plus

Nokia 8810



Nokia Mediamaster



Brief History of ARM

- ARM is the short form of **Advanced RISC Machines** Ltd.
- Founded 1990, owned by **Acorn**, Apple and VLSI
- Known as computer manufacturer before becoming ARM
- Acorn developed a 32-bit RISC processor for it's own use
Acorn Archimedes

ARM architecture

- 32-bit RISC-processor core (32-bit instructions)
- 37 pieces of 32-bit integer registers (16 available)
- Pipelined (ARM7: 3 stages)
- Cached (depending on the implementation)
- Von Neuman-type bus structure (ARM7), Harvard (ARM9)
- 8 / 16 / 32 -bit data types
- 7 modes of operation (usr, fiq, irq, svc, abt, sys, und)
- Simple structure -> reasonably good speed / power consumption ratio

ARM Architecture Versions

Popular ARM Architectures

ARM7

- 3 pipeline stages (fetch/decode/execute)
- High code density / low power consumption
- Most widely used for low-end systems

ARM9

- Compatible with ARM7
- 5 stages (fetch/decode/execute/memory/write)
- Separate instruction and data cache

ARM10

- 6-stages (fetch/decode/CO/ execute/memory/write)

ARM Architecture Versions

Naming ARM

- ARMxyzTDMIEJFS
 - x: series
 - y: MMU
 - z: cache
 - T: Thumb
 - D: debugger
 - M: Multiplier
 - I: EmbeddedICE (built-in debugger hardware)
 - E: Enhanced instruction
 - J: Jazelle (JVM)
 - F: Floating-point
 - S: Synthesizable version (source code version for EDA tools)

ARM Architecture Versions (continued...)

ARM Family Comparison

	ARM 7 (1995)	ARM9 (1997)	ARM10 (1999)	ARM11 (2003)
Pipeline depth	3-stage	5-stage	6-stage	8-stage
Typical clock frequency (MHz)	80	150	260	335
Power (mW/MHz)	0.06	0.19	0.50	0.40
Throughput (MIPS/MHz)	0.97	1.1	1.3	1.2
Architecture	Non Neumann	Harvard	Harvard	Harvard
Multiplier	8 x 32	8 x 32	16 x 32	16 x 32

3 pipeline stages (fetch/decode/execute)

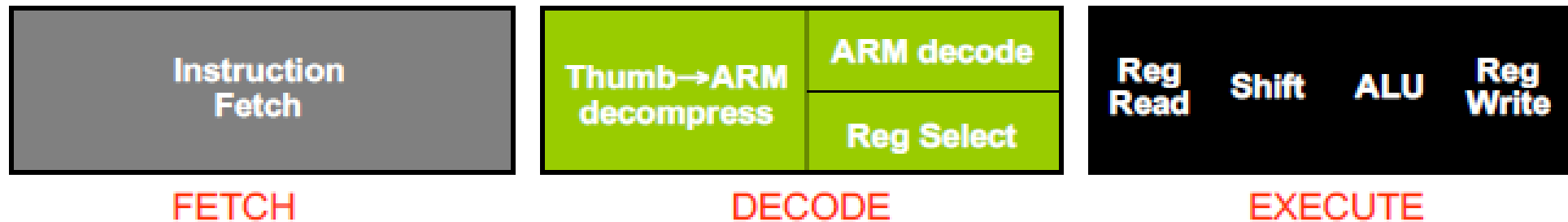
5 stages (fetch/decode/execute/memory access/write back)

6-stages (fetch/issue/decode/execute/memory access/write back)

ARM Architecture Versions (continued...)

ARM Core Pipelines

ARM7TDMI



ARM9TDMI



ARM Architecture Versions (continued...)

ARM is based on RISC Architecture

RISC supports simple but powerful instructions that execute in a single cycle at high clock frequency

Major design features:

- **Instructions:** reduced set / single cycle / fixed length
- **Pipeline:** decode in one stage / no need for microcode
- **Registers:** large number of general-purpose registers (GPRs)
- **Load/Store Architecture:** data processing instructions work on registers only; load/store instructions to transfer data from/to memory.

Now-a-days *CISC machines also implement RISC concepts*

ARM Architecture Versions (continued...)

ARM architecture is different from pure RISC:

- Variable cycle execution for certain instructions (multiple-register load/store for higher code density)
- In-line barrel shifter results in more complex instructions (improves performance and code density).
- Thumb 16-bit instruction set (results in improvement in code density by about 30%).
- Conditional execution (reduces branch and improves performance).
- Enhanced instructions (some DSP instructions are present).

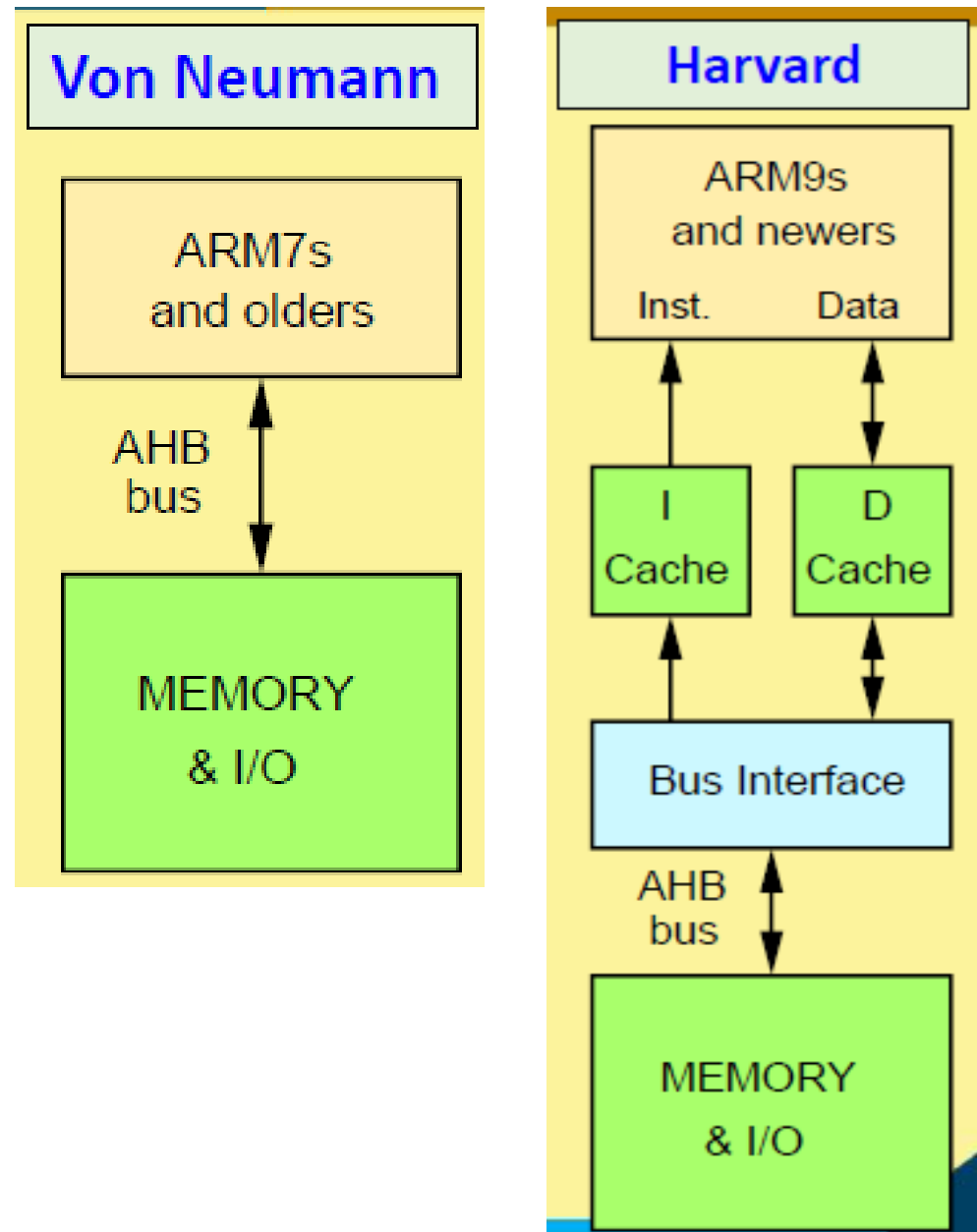
ARM Architecture Versions (continued....)

Memory-mapped I/O:

- No specific instructions for I/O
- Use Load/Store instr. for I/O
- Peripheral's registers at some memory addresses

AHB -

Advanced High-performance **Bus**



ARM Architecture Versions (continued...)

Architecture versions

Name	ARM instruction set version	Thumb instruction set version	Notes
ARMv4	4	None	-
ARMv4T	4	1	-
ARMv5T	5	2	-
ARMv5TExP	5	2	Enhanced DSP instructions except LDRD, MCRR, MRRC, PLD, and STRD
ARMv5TE	5	2	Enhanced DSP instructions
ARMv5TEJ	5	2	Addition of BXJ instruction and Jazelle Extension support over ARMv5TE

ARM Architecture Versions (continued....)

ISA Versions and Part Numbers

ARM Family	ARM Architecture	ARM Core
ARM1	ARMv1	ARM1
ARM2	ARMv2	ARM2
	ARMv2a	ARM250
ARM3	ARMv2a	ARM3
ARM6	ARMv3	ARM60
		ARM600
		ARM610
ARM7	ARMv3	ARM700
		ARM710
		ARM710a

ARM Architecture Versions (continued....)

ISA Versions and Part Numbers

ARM Family	ARM Architecture	ARM Core
ARM1	ARMv1	ARM1
ARM2	ARMv2	ARM2
	ARMv2a	ARM250

What is the difference between the ARM architecture and the ARM core?

ARM6	ARMv3	ARM600
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Core - A processor interfaced with essential components (such as Flash, RAM, ROM) and feature components (Capsense, BLE, I2C etc)

		ARM710a
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ARM Architecture Versions (continued....)

ISA Versions and Part Numbers (cont)

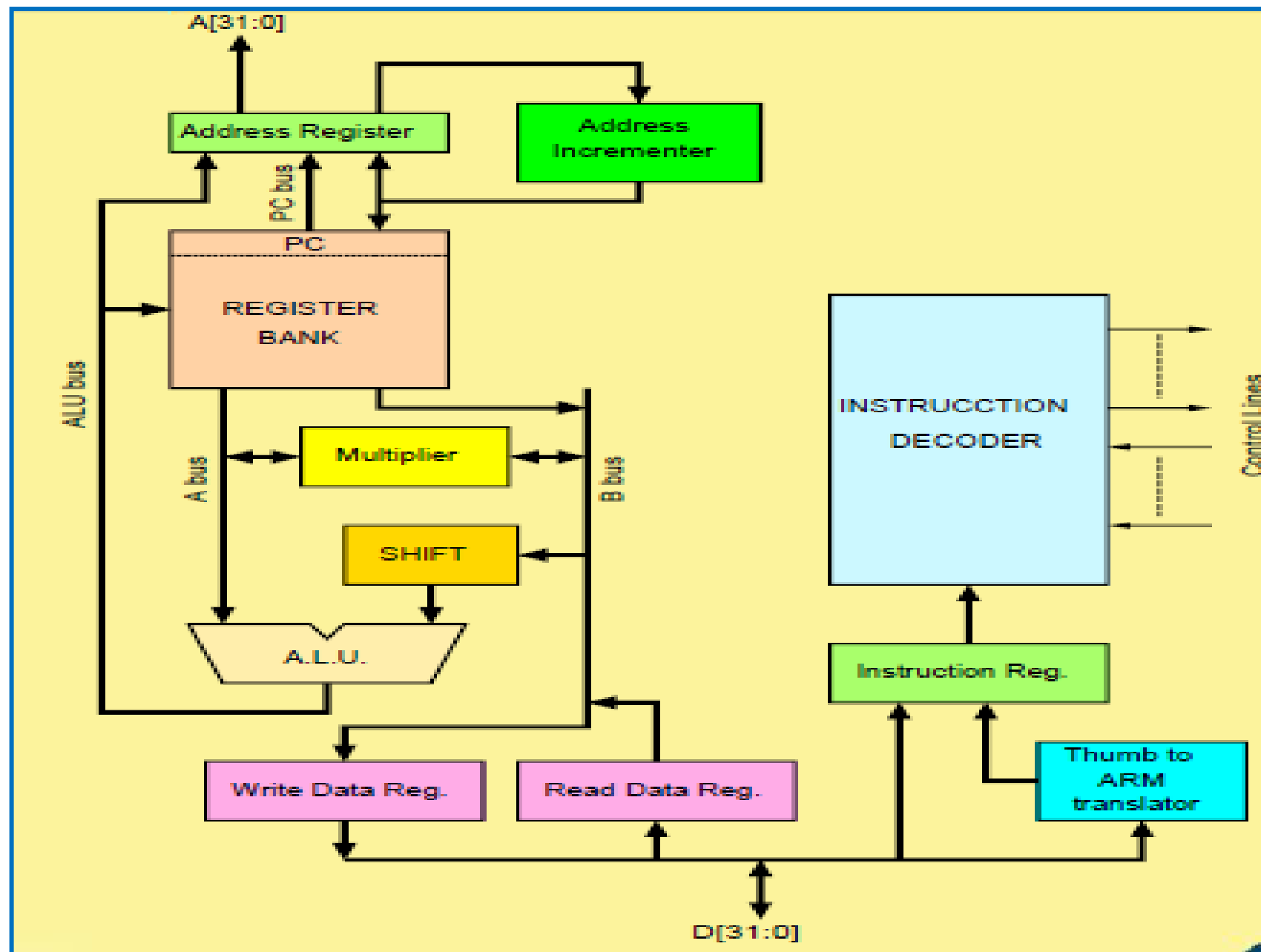
ARM Family	ARM Architecture	ARM Core
ARM7TDMI → ARMv4T		ARM7TDMI(-S)
		ARM710T
		ARM720T
		ARM740T
Version 4 and the introduction of Thumb (T variant)		
ARM7EJ →	ARMv5TEJ	ARM7EJ-S
Version 5 ; T variant; Enhanced DSP instructions and Jazelle Extension		
StrongARM	ARMv4	SA-1
ARM9TDMI	ARMv4T	ARM9TDMI
		ARM920T
		ARM922T
		ARM940T

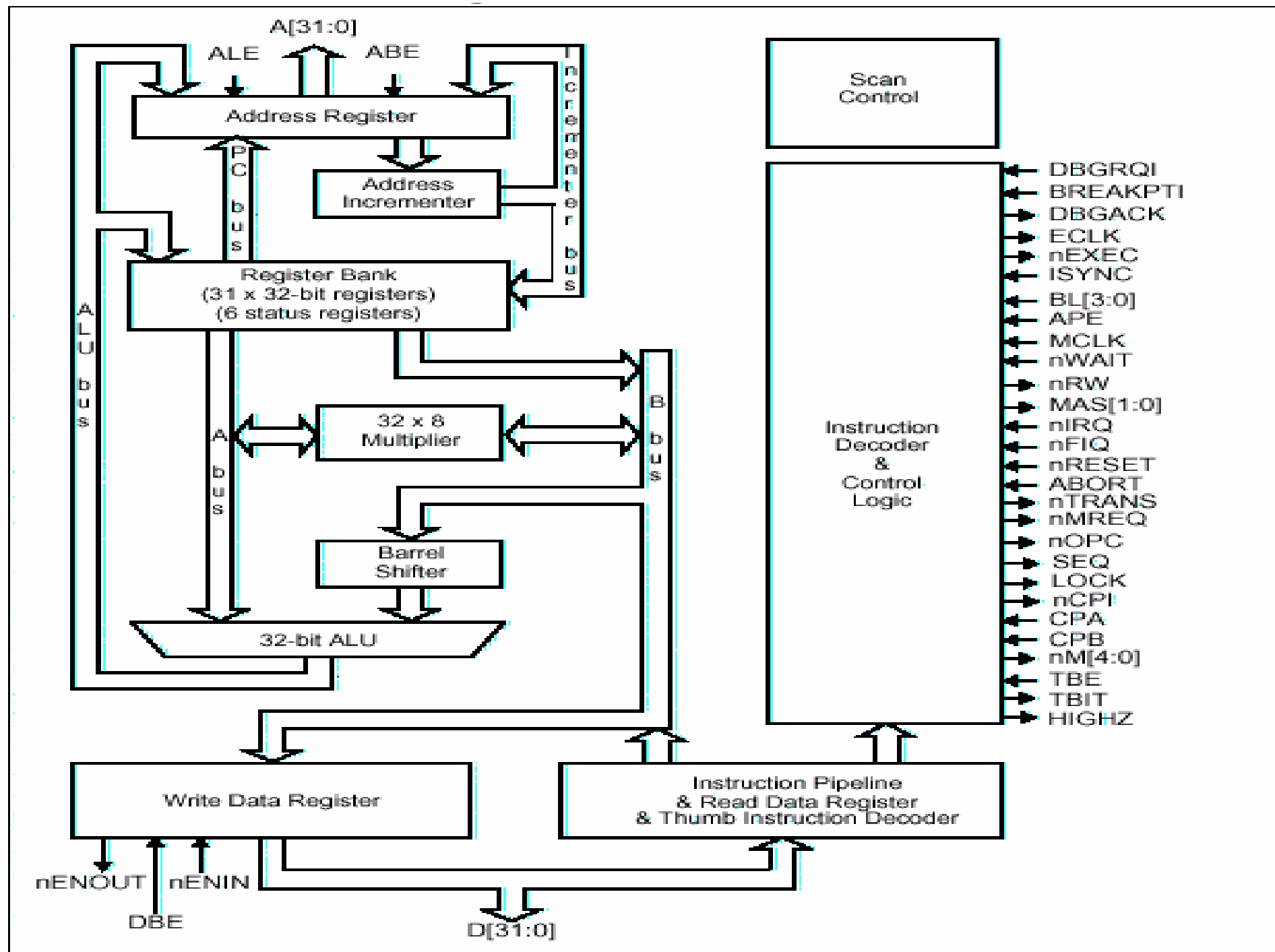
ARM Architecture Versions (continued...)

ARM7TDMI Processor Core

- ❑ Current low-end ARM core for applications like digital mobile phones
- ❑ TDMI
 - **T**: Thumb, 16-bit compressed instruction set
 - **D**: on-chip Debug support, enabling the processor to halt in response to a debug request
 - **M**: enhanced Multiplier, yield a full 64-bit result, high performance
 - **I**: Embedded ICE hardware
- ❑ Von Neumann architecture
- ❑ 3-stage pipeline, CPI ~ 1.9

ARM Architecture - Typical ARM7 Architecture





About the ARM architecture (continued...)

Architecture Features

- Large uniform register file
- Load/store architecture (only regr content not mem)
- Simple addressing modes (Regrs or Instn fields)
- Uniform and fixed-length instruction fields, to simplify instruction decode.

About the ARM architecture (continued...)

Architecture Features (continued...)

- The ARM architecture provides - control over both the Arithmetic Logic Unit (ALU) and shifter in most data-processing instructions
- Auto-increment and auto-decrement addressing modes to optimize program loops
- Load and Store Multiple registers to maximize data throughput
- Conditional execution of almost all instructions to maximize execution throughput.

About the ARM architecture (continued...)

ARM processor modes and registers

Processor Mode	Code	Description
User	usr	Normal program execution mode
FIQ	fiq	Entered when a high priority (fast) interrupt is raised
IRQ	irq	Entered when a low-priority (normal) interrupt is raised
Supervisor	svc	A protected mode for the operating system (entered on reset and when software interrupt instruction is executed)
Abort	abt	Used to handle memory access violations
Undefined	und	Used to handle undefined instructions
System	sys	Runs privileged operating system tasks

About the ARM architecture (continued...)

ARM Registers

ARM has 37 registers all of which are 32-bits long

These registers are:

- 1 dedicated program counter (PC)
- 1 dedicated current program status register (CPSR)
- 5 dedicated saved program status registers (SPSR)
- 30 general-purpose registers (GPR)

The current processor mode governs which of the several register sets is accessible.

About the ARM architecture (continued...)

ARM Registers (contd....)

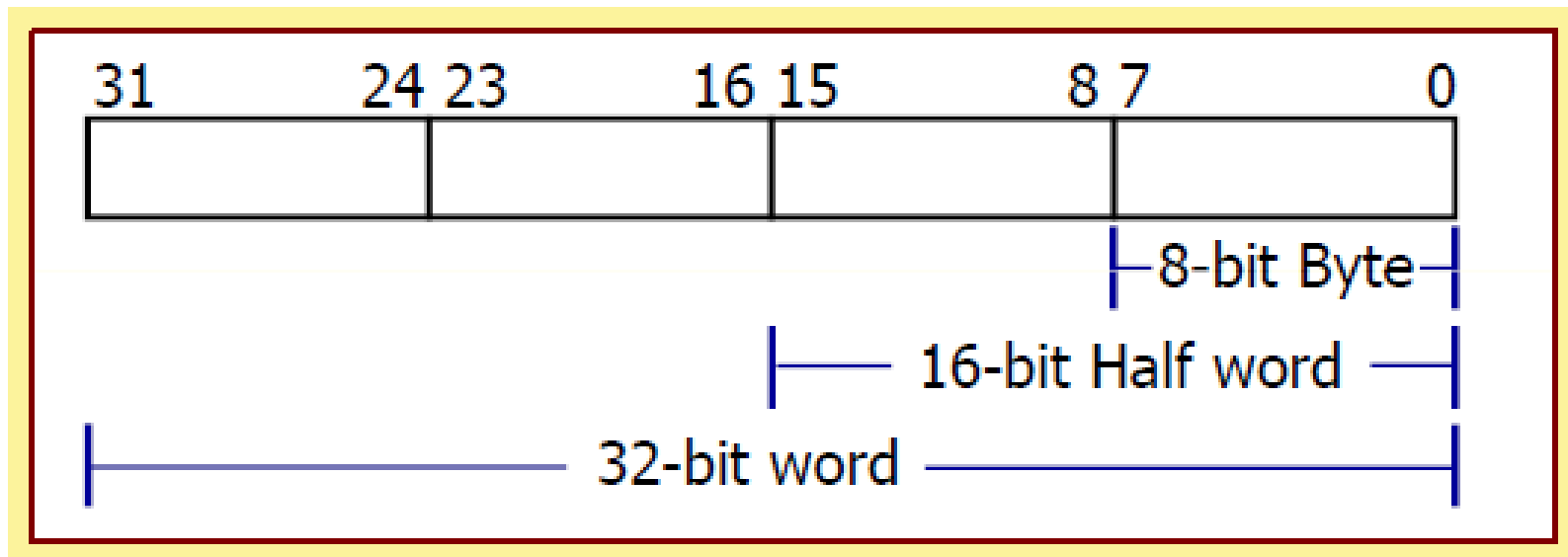
Only 16 registers are visible to a specific mode of operation.
Each mode can access:

- A particular set of registers r0-r12
- r13 (SP, stack pointer)
- r14 (LR, link register) (return addr after subroutine call or exceptions)
- r15 (PC, program counter) (points to - two instructions after the instruction currently being executed)
- Current program status register (CPSR)
- Privileged modes (except System) can also access a particular SPSR

About the ARM architecture (continued...)

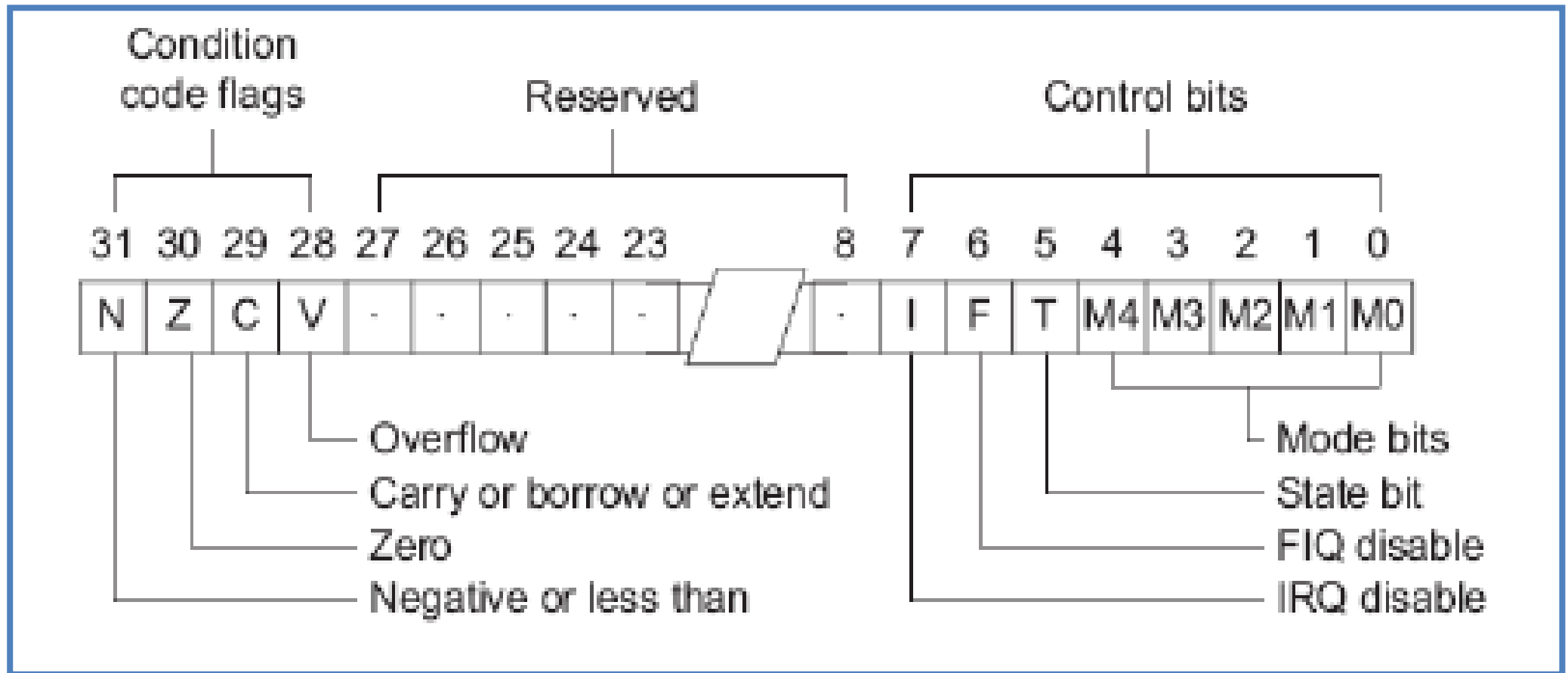
General-purpose Registers

- 6 data types are supported (signed/unsigned)
8-bit byte, 16-bit half-word, 32-bit word
- All ARM operations are 32-bit.
 - Shorter data types are only supported by data transfer operations



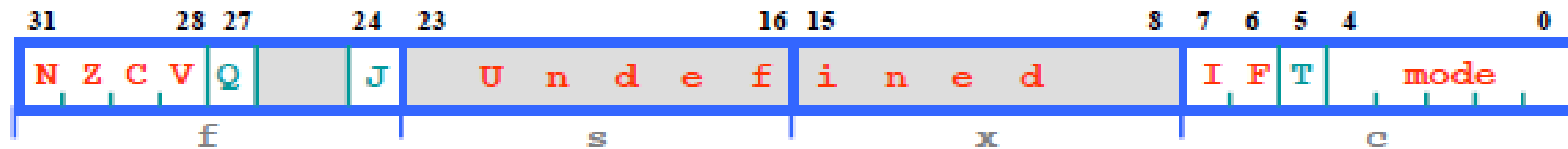
About the ARM architecture (continued...)

Current Program Status Register (CPSR)



About the ARM architecture (continued...)

Current Program Status Register (CPSR)



■ Condition code flags

- N = Negative result from ALU
- Z = Zero result from ALU
- C = ALU operation Carried out
- V = ALU operation oVerflowed

■ Sticky Overflow flag - Q flag

- Architecture 5TE/J only
- Indicates if saturation has occurred

■ J bit

- Architecture 5TEJ only
- J = 1: Processor in Jazelle state (processor executes Java Bytecode)

■ Interrupt Disable bits.

- I = 1: Disables the IRQ.
- F = 1: Disables the FIQ.

■ T Bit

- Architecture xT only
- T = 0: Processor in ARM state
- T = 1: Processor in Thumb state

■ Mode bits

- Specify the processor mode

About the ARM architecture (continued...)

Current Program Status Register (CPSR)

Mode bits (M4–M0) determines the processor operating modes

Mode Bit settings (M4–M0)

USER mode:	10000b	SYS mode:	11111b
FIQ mode :	10001b	IRQ mode:	10010b
SVC mode:	10011b	ABORT mode:	10111b
UNDEFINE mode:	1100b		



About the ARM architecture (continued...)

Special Registers

- PC (r15): Any instruction with PC as its destination register is a program branch.
- LR (r14): Saves a copy of PC when executing the BL instruction (subroutine call) or when jumping to an exception or an interrupt handler.
 - It is copied back to PC on return from those routines
- SP (r13): There is no stack in the ARM architecture
 - R13 is reserved as a pointer for the software-managed stack

About the ARM architecture (continued...)

Special Registers (continued....)

CPSR: Holds the visible status register.

SPSR: Holds a copy of the previous status register while executing exception or interrupt handler routines.

- Copied back to CPSR on return from exception or interrupt.

About the ARM architecture (continued...)

Program Counter

- When the processor is executing in ***ARM mode / state***:
 - ✓ All instructions are 32-bits wide, and must be word aligned.
 - ✓ The last two bits of PC are zero (i.e. not used).
 - ✓ Due to pipelining, PC points 8 bytes ahead of the current instruction, or 12 bytes ahead if the current instruction includes a register-specified shift.
- When the processor is executing in ***Thumb mode / state*** :
 - ✓ All instructions are 16-bits wide, and are half-word aligned.
 - ✓ The last bit of the PC is zero (i.e. not used).

About the ARM architecture (continued...)

Processor Operating States

- **ARM state** – Executes 32bit word aligned ARM instructions
- **Thumb state** on T variants - Executes 16 bit half word aligned Thumb instructions [the PC can be half word (16-bit) in this state]
- **Jazelle state** on J variants – Jazelle cores can also execute Java bytecode [the PC can be byte aligned in this state]

ARM7TDMI Core Diagram

